La Scuola d'Italia / Dr. Huson / IB Math (b)
17 April 2026

First & Last name:

6.8 Quiz: Polynomials (no calculator)

1. [Maximum mark: 8]

   Let $f(x) = x^2 + 6x - 16$.

   (a) Write $f(x)$ in factored form, such that $f(x) = (x - a)(x - b)$. [2]

   (b) Write down the x-intercepts of the graph of f. [1]

   (c) Write down the y-intercept. [1]

   (d) Write $f(x)$ in vertex form, i.e. $f(x) = (x - h)^2 + k$. [3]

   (e) Write down the coordinates of the vertex of the parabola. [1]

2. [Maximum mark: 7]

Consider the function $g(x) = -(x-5)^2 + 4$.

(a) Write down the coordinates of the vertex. [2]

(b) Write down the equation of the axis of symmetry. [1]

(c) Show that $g(x) = -x^2 + 10x - 21$. [2]

(d) Solve $g(x) = 0$ to find the x-intercepts. [2]

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3. [Maximum mark: 6]

The line $y = 4x$ and the parabola $y = x^2 - 5$ are drawn on the same axes.

(a) Show that the x-coordinates of the points of intersection satisfy
$x^2 - 4x - 5 = 0$. [1]

(b) Show that $x^2 - 4x - 5 = 0$ has two distinct real roots. [1]

(c) Factorise $x^2 - 4x - 5$. [1]

(d) Find the x-coordinates of the points of intersection. [1]

(e) Find the coordinates of both points of intersection as (x, y) pairs. [2]

4. [Maximum mark: 7]

The equation $2x^2 - kx + 32 = 0$ has two equal roots.

(a) Write down the discriminant in terms of k. [1]

(b) Find the values of k. [2]

(c) For the positive value of k, find the repeated root. [2]

(d) State the values of k for which the equation has no real roots. [2]